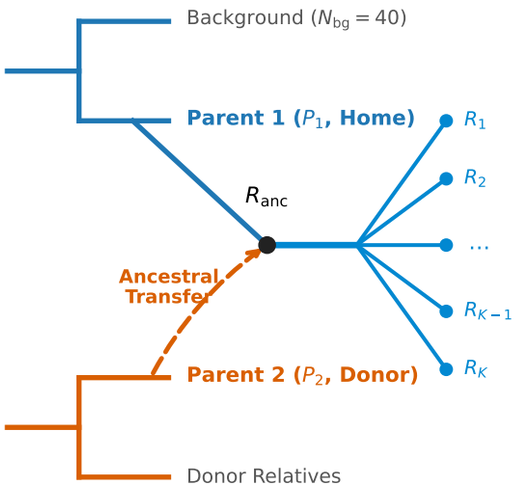
A. Epidemic Clonal Expansion (Scenario 1.1)



Descendant Swarm ($K \in \{2, 5, 8, 10, 15\}$)

- True physical events: $K_{true} = 1$
- Pairwise Triplet Tests (3SEQ): $\hat{K} = 10$--55 false events
- RhizAeon Bundle Tracking: $\hat{K} = 1.00$ (100% exact)

Shared Recombinant Genome ($L = 5{,}000$ nt) — Identical cassette across all K isolates:



B. Multi-Way Mosaic Swarms (Scenario 1.2)

1. Dual Cassette ($K_{donor} = 2$ donors, 2 separate non-overlapping tracts)



RhizAeon: 2.00 / 2.00 donors resolved (100%) | 3SEQ: 1.02 donors (misses P_3)

2. Nested Russian Doll ($K_{donor} = 2$ donors, hierarchical embedded introgression)



RhizAeon: 1.10 donors resolved | 3SEQ: 1.00 donor (completely blind to inner P_3)

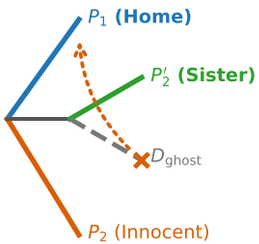
3. Triple Crossover Cascade ($K_{donor} = 2$ transitions, serial chromosome mosaic)



RhizAeon: 1.63 donors resolved, 100% recall | 3SEQ: 95.3% background parent false alarms

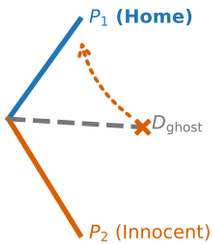
C. Ghost Donor Introgression (Scenario 1.3)

(i) Surrogate Sister



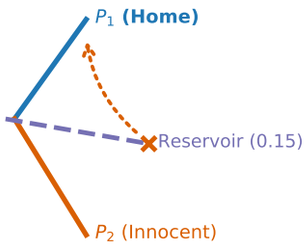
True donor unsampled.
Sister clade P_2' present.
- Goal: Attribute to P_2'
- Zero false calls on P_2

(ii) Pure Ghost Lineage



Deep ancestral lineage.
No donor relatives sampled.
- Goal: Open-world bounds
- Zero false calls on P_2

(iii) Deep Reservoir



Cross-species reservoir.
High divergence ($d = 0.15$).
- Goal: Open-world bounds
- Zero false calls on P_2

• Triplet Trap (3SEQ): 100% false framing of innocent P_2
• RhizAeon Open-World: 98.3% sister recall, 0.0% false framing

D. Bacterial Micro-Conversions (Scenario 1.4)

Mosaic Structure	Tract Length	Mutations	Power (Rhiz / 3SEQ)
	800 nt	16--74 muts	**100% / 100%**
	400 nt	7--38 muts	**98% / 100%**
	200 nt	4--20 muts	**91% / 96%**
	100 nt	2--9 muts	**89% / 55%**
	50 nt	1--5 muts	**40% / 5%**

1 nt 5,000 nt

Continuous Dyadic Scale-Space Matched Filter (ψ_τ, $\tau \in \{15, 25, 50, 100, 200\}$ codons):



Wavelet localization achieves median breakpoint error of 13--19 nt